BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

ES C221 MECHANICS OF SOLIDS

ASSIGNMENT: 8 (DEFLECTIONS)

1) A cast iron beam 40 mm wide and 80 mm deep is simply supported on a span of 1.2 m. The beam carries a point load of 15 kN at the centre. Find the deflection at the centre. Take E = 108000 N/mm$^2$. When the beam is subjected to uniformly distributed load of w/m, what is the deflection at the centre of the beam? **[y= 2.3 mm; y=** $-\frac{5wl^4}{384EI}$ **]**

2) A horizontal beam of symmetrical rectangular section simply supported at the ends, carries a load whose intensity varies uniformly from 10 kN/m at one end to 50 kN/m at the other. Find the central deflection if the span is 9 m, the section is 500 mm deep and the maximum bending stress is 80 N/mm$^2$. Take E = 210 kN/mm$^2$. **[y=12.7 mm]**

3) A timber cantilever of rectangular section is 75 mm wide and 250 mm deep at the fixed end and tapers uniformly to 75 mm wide and 100 mm deep at the free end. The projecting length is 1800 mm and there is a load of 1800 N at the free end. Calculate the deflection of the free end and the maximum bending stress. Take E = 14000 N/mm$^2$. **[4.84 mm; 6.48 N/mm$^2$]**

4) Using double integration method, calculate the slope and deflection for the following cases:

a) Slope and deflection at free end of Cantilever beam with uniformly varying intensity (o at free end and w/m at fixed end) $[\theta = \frac{wl^3}{24EI}; y = -\frac{wl^4}{30EI}]$

b) Slope and deflection at free end of cantilever beam of uniformly distributed load on a part of span from fixed end $[\theta = -\frac{wa^3}{6EI}; y = -\frac{wa^4}{8EI} - \frac{wa^3}{6EI}(l-a)]$

c) Maximum deflection for the simply supported beam with distributed load of varying intensity (o at one end and w/m at other end) $[y_{max} = -0.00652\frac{wl^4}{EI}]$

5) A cantilever of length 2 metres carries a uniformly distributed load of 2500 N/m for a length of 1.25 m from the fixed end and a point load of 1000 N at the free end. If the section is rectangular 120 mm side and 240 mm deep, find the deflection at the free end. Take E = 10000 N/mm$^2$ **[y = 2.922 mm]**

6) Find the central deflection of the uniform bend shown in Fig Q.6. $[\frac{11pa^3}{6EI}]$

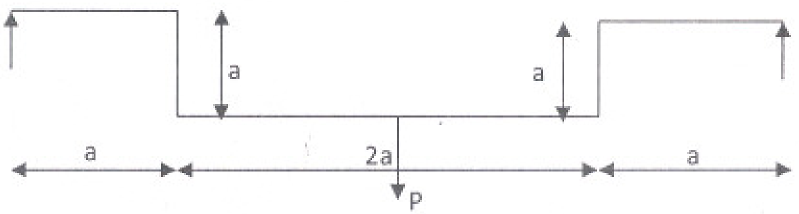
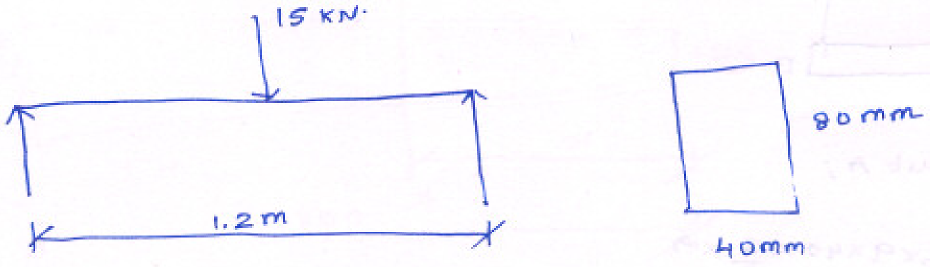



Fig. Q6

ASSIGNMENT-8 (DEFLECTIONS)

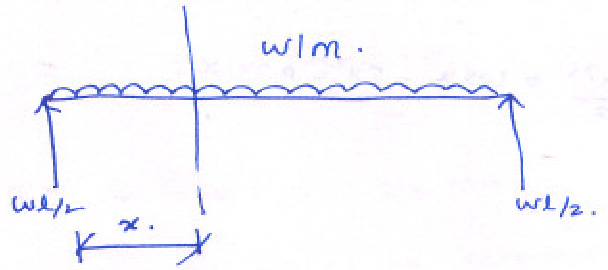
SOLUTIONS:

①




$$y_c = \frac{Wl^3}{48EI} = \frac{15 \times (1.2)^3}{48 \times 108 \times 10^{-6} \times \frac{40 \times 80^3}{12}} = \boxed{2.3\text{ mm};}$$

b).




$$EI \frac{d^2y}{dx^2} = \frac{wl}{2}x - \frac{wx^2}{2} \rightarrow (1)$$

$$EI \frac{dy}{dx} = \frac{wl}{4} \cdot x^2 - \frac{wx^3}{6} + C_1 \rightarrow (2)$$

Loading → symmetrical; Max def @ mid span;

At $x = L/2 \Rightarrow \frac{dy}{dx} = 0;$

$$(2) \Rightarrow \quad 0 = \frac{wl}{4}\left(\frac{L}{2}\right)^2 - \frac{w}{6}\left(\frac{L}{2}\right)^3 + C_1$$

$$\Rightarrow \quad C_1 = \frac{-wl^3}{24};$$

$$\therefore EI\frac{dy}{dx} = \frac{wl}{4}x^2 - \frac{wx^3}{6} - \frac{wl^3}{24}$$ [slope equation].

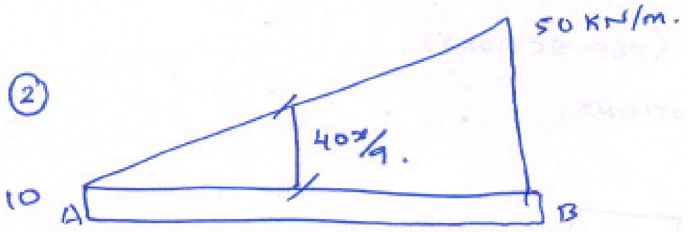
$$EIy = \frac{wl}{12}x^3 - \frac{wx^4}{24} - \frac{wl^3}{24}x + C_2$$

At $x = 0;\ y = 0. \Rightarrow C_2 = 0;$

$$EIy = \frac{wl}{12}x^3 - \frac{wx^4}{24} - \frac{wl^3 x}{24}$$ (Deflection equation).

At $\cancel{x} = l/2 \Rightarrow \boxed{y = \frac{-5wl^4}{384EI}}$

(2)




Taking moments about A;

$R_B \times 9 = 10 \times 9 \times 4.5 + \frac{1}{2} \times 9 \times 40 \times \frac{2}{3} \times 9$

$R_B = 165$ KN; $R_A = 105$ KN;

At a distance $x$ from 9;

Shear force: $105 - 10x - \frac{1}{2}x \cdot \frac{40x}{9} = 105 - 10x - 2.222x^2$

Bending moment $= 105x - \frac{10x^2}{2} - 2.222\frac{x^3}{3} = 105x - 5x^2 - 0.741x^3$.

Max. B.M occurs @ zero S.F;

$\boxed{x = 4.983 m}$

$\therefore$ Max Bending moment $= 307.6$ KN.m;

Max Bending stress $= 80 = \frac{307.6 \times 10^6}{I} \times \frac{500}{2} \Rightarrow I = 961.25 \times 10^6 \; mm^4$

$EI\frac{d^2y}{dx^2} = 105x - 5x^2 - 0.741x^3$.

$EI\frac{dy}{dx} = 105.\frac{x^2}{2} - 5\frac{x^3}{3} - 0.741\frac{x^4}{4} + C_1$

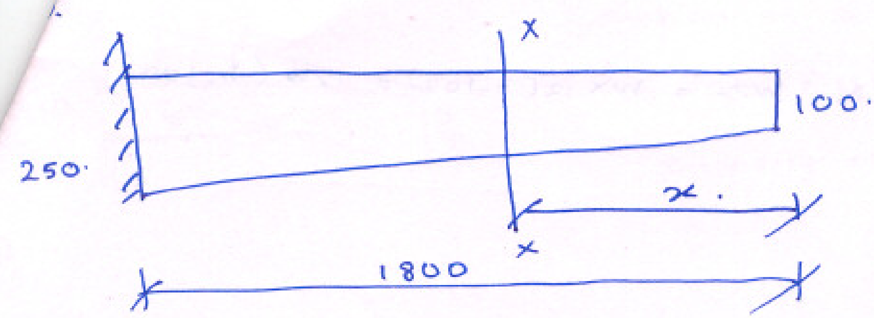
$EIy = 105.\frac{x^3}{6} - 5\frac{x^4}{12} - 0.741\frac{x^5}{20} + C_1x + C_2$

At $x = 0$; $y = 0$; $C_2 = 0$;

At $x = 9m$; $y = 0 \Rightarrow$. $C_1 = -870.7$ KN/m$^2$.

$\therefore EI y_{(centre)} = -2562.7$ KN/m$^2$.

$\boxed{y = 12.7 \text{ mm}}$




At any section X distant $x$ from free end;

Depth of section $h = 100 + \left(\frac{250-100}{1800}\right)x$ $\quad \therefore h = 100 + \frac{x}{12}.$

$x = 12(h-100).$

M.I $\quad I_{xx} = \frac{75 \times h^3}{12} = \frac{25 h^3}{4}\ mm^4.$

B.M @ section $\quad M = W.x$

Strain energy stored by cantilever:

$$W_i = \int \frac{M^2\,dx}{2E I_{xx}} = \int_0^{1800} \frac{W^2 \cdot x^2 \cdot dx}{2E \cdot \frac{25h^3}{4}} = \frac{2W^2}{25E}\int_0^{1800} \frac{x^2}{h^3} \cdot dx$$

$$W_i = \frac{2W^2}{25E}\int_{100}^{250} \frac{144(h-100)^2 \cdot \times 12\,dh}{h^3} = \frac{3456}{25}\,\frac{W^2}{E}\int_{100}^{250} \frac{(h^2 - 200h + 10000)}{h^3}\,dh.$$

$$W_i = \frac{3456}{25} \cdot \frac{W^2}{E}\left[\log h + \frac{200}{h} - \frac{5000}{h^2}\right]_{100}^{250}$$

$$W_i = \frac{3456}{25}\,\frac{W^2}{E}\left[\log 250 + \frac{200}{250} - \frac{5000}{250^2}\right] - \left(\log 100 + \frac{200}{100} - \frac{5000}{100^2}\right).$$

$$\therefore W_i = \frac{3456}{25} \cdot \frac{W^2}{E}\left[\log\frac{5}{2} - \frac{39}{50}\right]$$

$$\frac{1}{2} \cdot W \cdot \delta =$$

$$\delta = \frac{6912}{25}\left(\log\frac{5}{2} - \frac{39}{50}\right)\cdot\frac{W}{E}$$

$$\boxed{\delta = 4.84\ mm;}$$

B.M @ any section :- $M = W.x = W \times 12(h-100) = 12W(h-100).$

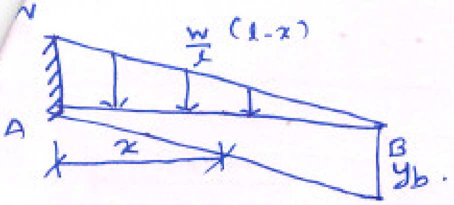
$$Z = \frac{75h^2}{6} = \frac{25}{2}h^2.$$

$$f = \frac{36W}{25}\left(\frac{1}{h} - \frac{100}{h^2}\right).$$

$$\frac{df}{dh} = 0 \Rightarrow \frac{df}{dh} = \frac{36W}{25}\left(-\frac{1}{h^2} + \frac{200}{h^3}\right) = 0.$$

$h = 200\ mm;$

$$\boxed{f_{max} = \frac{36}{25} \times 1800\left(\frac{1}{200} - \frac{100}{200^2}\right) = 6.48\ N/mm^2}$$




Consider a section X at a distance $x$ from fixed end A;

Intensity of loading at $x$;

$$= \left(\frac{l-x}{l}\right) \times w./m.$$

a)

B.M at section $x$ is given by.

$$EI.\frac{d^2y}{dx^2} = -\frac{1}{2}(l-x).\frac{w}{l}(l-x).\frac{(l-x)}{3}$$

$$= \frac{-w(l-x)^3}{6l}$$

$$EI.\frac{dy}{dx} = \frac{w(l-x)^4}{24l} + C_1$$

At A, slope is zero;

At $x=0$; $\frac{dy}{dx}=0$; $0 = \frac{wl^3}{24} + C_1 \Rightarrow C_1 = -\frac{wl^3}{24}$;

$$EI.\frac{dy}{dx} = \frac{w(l-x)^4}{24l} - \frac{wl^3}{24} \quad \text{(slope equation)}$$

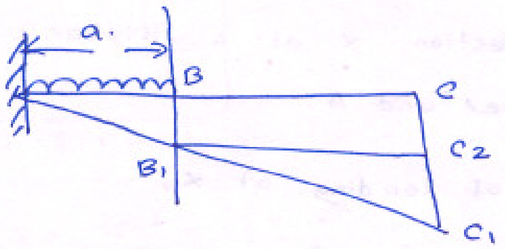
$$EI.y = -\frac{w(l-x)^5}{120l} - \frac{wl^3}{24}x + C_2$$

$x=0$; $y=0$; $0 = -\frac{wl^4}{120} + C_2 \Rightarrow C_2 = \frac{wl^4}{120}$;

$$EIy = -\frac{w(l-x)^5}{120l} - \frac{wl^3}{24}x + \frac{wl^4}{120} \quad \text{.... (Def. equation)}$$

$$\boxed{\theta = -\frac{wl^3}{24EI}; \quad y = -\frac{wl^4}{30EI}}$$

b)




Deflection at $B = y_b = BB_1 = \frac{wa^4}{8EI}$;

Let $B_1C_2$ ⊥r to $CC_1$.

$$CC_2 = BB_1 = y_b = \frac{wa^4}{8EI};$$

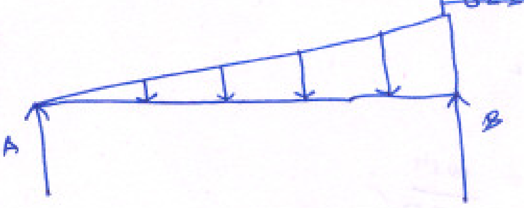
slope at $B = \theta_B = \frac{wa^3}{6EI}$;

$$C_2C_1 = B_1C_2\,\theta_B = (l-a)\cdot\frac{wa^3}{6EI}$$

∴ Deflection at $C = y_c = CC_1 + C_2C_1 = \boxed{\frac{wa^4}{8EI} + (l-a)\frac{wa^3}{6EI}}$.

The portion of cantilever from C to B will remain straight having a slope of $\boxed{\frac{wa^3}{6EI}}$;

c).




Intensity of load = $\frac{wx}{l}$.

Taking moments about B,

$$R_A \times l = \frac{wl}{2} \times \frac{l}{3}$$

$$R_A = \frac{wl}{6};\quad R_b = \frac{wl}{3}$$

B.M @ any section at a distance $x$ from A;

$$= \frac{wl}{6}x - \frac{wx}{l}\cdot\frac{x}{2}\cdot\frac{x}{3} = \frac{wl}{6}x - \frac{wx^3}{6l}$$

$$EI\frac{d^2y}{dx^2} = \frac{wlx}{6} - \frac{wx^3}{6l}.$$

$$EI\frac{dy}{dx} = \frac{wlx^2}{12} - \frac{wx^4}{24l} + C_1$$

$$EI\,y = \frac{wlx^3}{36} - \frac{wx^5}{120l} + C_1x + C_2.$$

At $x=0$; $y=0$; $C_2 = 0$;

At $x=l$; $y=0$; $C_1 = \frac{-7wl^3}{360}$;

$$EIy = \frac{wlx^3}{36} - \frac{wx^5}{120l} - \frac{7wl^3}{360}x$$

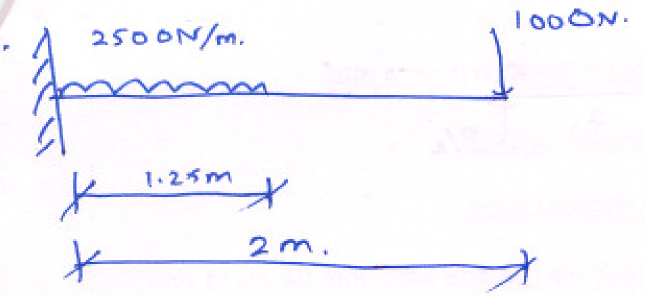
To find max def, equate slope $=0$;

$$EI\frac{dy}{dx} = \frac{wlx^2}{12} - \frac{wx^4}{24l} - \frac{7wl^3}{360}$$

$\Rightarrow x = 0.5193l$;

$$EIy = \frac{wl(x=)^3}{36} - \ldots\ldots$$

$$\boxed{y = -\frac{0.00652\,wl^4}{EI}}$$



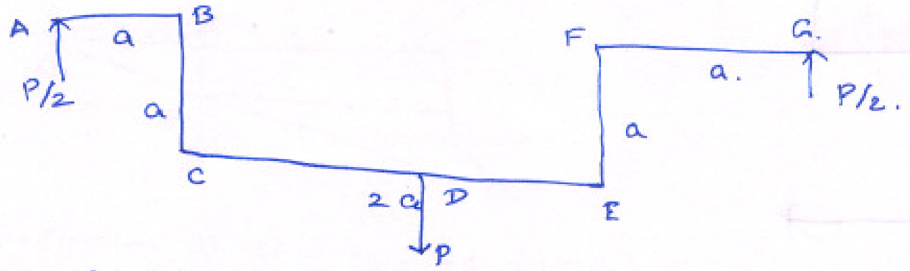

$$\delta = \frac{wa^3}{8EI} + \frac{wa^3}{6EI}(l-a). \quad \text{[ due to uniformly distributed ]}$$

$$= \frac{1373}{EI};$$

$$\delta = \frac{2667}{EI} \left[ \frac{Wl^3}{3EI} \right] \quad \text{( due to point load )}$$

$$\boxed{\text{Net deflection @ C = 2.92 mm.}}$$

6).




$$W_i = \Sigma \int \frac{M^2 ds}{2EI}$$

$$= 2\left[ \int_0^a \frac{\left(\frac{P}{2} \cdot x\right)^2 dx}{2EI} + \int_0^a \frac{\left(\frac{P}{2} \cdot a\right)^2 dy}{2EI} + \int_0^a \left(\frac{P}{2} \cdot (x+a)\right)^2 \frac{dx}{2EI} \right.$$

$$= 2\left[ \frac{P^2}{4} \cdot \frac{a^3}{3} \cdot \frac{1}{2EI} + \frac{P^2}{4} \cdot \frac{a^2}{2EI} \cdot a + \frac{1}{2EI} \cdot \frac{1}{3} \cdot \left((x+a)^3\right)_0^a \right]$$

$$= 2\left[ \frac{P^2 a^3}{24EI} + \frac{P^2 a^3}{8EI} + \frac{P^2}{24EI}\left(8a^3 - a^3\right) \right]$$

$$W_i = \frac{11}{12} \cdot \frac{P^2 a^3}{EI.}$$

$$\boxed{\delta_v = \frac{\partial W_i}{\partial P} = \frac{11}{6} \frac{P a^3}{EI}}$$